



DPUSNDXEgoWAV

DPUSNDXEgoWAV simulates engine and wind/roll noises of a car by using several WAVE files. This DPU is an alternative to DPUSNDXEgo which uses sample data of a special format (SIM-files).

1. Inputs:

<i>RPM</i>	engine speed [500..7000]
<i>AcceleratorPedal</i>	position of the accelerator pedal [0..1]
<i>v_kmh</i>	vehicle speed [0..255]
<i>Horn</i>	horn (0/1)
<i>Indicator</i>	indicator relais (0/1)
<i>Starter</i>	ignition (0/1)
<i>Rumpeln</i>	rumble (or other)

2. Parameters:

<i>AudioData</i>	directory below lib/sounds/ where the samples are located (default: „bmw5“)
<i>VolStep</i>	change rate of volume during sample change (default: 0.05)
<i>TimeStep</i>	minimum time in ms before changing to another sample (default: 200)
<i>PitchFactor</i>	amplifier of pitch effect (default: 1)
<i>VolMotor</i>	engine sound volume (default: 1)
<i>VolWind</i>	wind sound volume (default: 1)
<i>VolMisc</i>	other sounds volume (default: 1)
<i>Motor[X,Y,Z]</i>	position of engine sound (default: [1.0,0.0,-0.5])
<i>Wind[X,Y,Z]</i>	position of wind sound (default: [0.5,0.0,0.5])
<i>Misc[X,Y,Z]</i>	position of other sounds (default: [1.0,0.0,-0.5])

3. Outputs:

<i>MSampleRPM</i>	idle engine speed of current sample
<i>MSampleLast</i>	load engine speed of current sample
<i>WSampleV</i>	wind speed of the current sample
<i>NSamplesRumpeln</i>	number of additional (rumble) samples loaded

4. Dependencies:

Several WAVE files are needed for the sound simulation. The following files will be loaded (some are mandatory):

/motor/rXXXXIYYY.wav



Engine sound for the speed of XXXX rounds per minute and engine load of YYY, for example r2350l050.wav should be a recording of an engine at 2350 rpm and 50% load (position of accelerator pedal).

/wind/vXXX.wav

Wind/roll sound for speed XXX, for example v100.wav should be a recording of the wind/roll sounds in the vehicle at 100 km/h.

/road/rollX.wav

Additional roll sound for different road types (i.e. concrete, cobble).

/misc/anlasser.wav

Sound of the ignition.

/misc/rumbleXX.wav

A rumble or other sample (may be omitted). These are loaded in alphabetical order and can be accessed by its ordinal number. For simplicity, you can name them rumble01.wav, rumble02.wav, rumble03.wav, and so on. That way, the number of the sample corresponds with the value the input must be set to, to play the file back.

/misc/hupe.wav

Sound of the horn.

/misc/blinker.wav

Sound of the indicator relais.

5. Functional principle:

DPUSNDXEgo loads sound samples for engine, wind/roll and other sounds and mixes them according to the inputs and parameters before playing them over OpenAL. To avoid cracking noises during the transition between two samples two sound sources are created of which usually only one is active. When changing samples both sources are active while fading the old sample out and the new sample in by change of volume. Every simulation step the volume is changed by the rate of *VolStep* until a volume of 1 or 0 respectively is reached. Using big values there might be noticable cracking sounds while using too small values delays the playback. The samples of the engine sound are additionally pitched in dependency of the engine speed. In the case the current engine speed is 2500 rpm and there is a sample available recorded for 2350 rpm the pitch is calculated as $2500/2350 * PitchFactor$.

Setting the input of *Starter* to 1 triggers playback of the ignition sound. Rumble (or any other sound) can be played by setting the *Rumpeln* input. The sound will be played as long as the value of the input is kept. As long as the input of *Horn* is set to 1 the horn sound will be played, same goes for the indicator relais sound if *Indicator* input is set to 1. As only one sound source is available for all other sounds (i.e. excluding engine and wind) the inputs are prioritized. With *Starter* set to 1 or the ignition



SILAB DRIVING SIMULATION

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sound playing all other inputs are ignored. *Rumpeln* is overruled by *Indicator* and both are overruled by *Horn*.